



EEHC DISTRIBUTION MATERIALS SPECIFICATION

EDMS 19- 400 - 2

Date: 3-08-2021

**EDMS 19-400-2
TECHNICAL SPECIFICATION**

FOR

High Rupture Capacity MV Current limiting Fuses

(HRC) 12KV & 24 KV

Issue: Aug-2021/ Rev- 2



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❖ توجد بنود اختيارية Option items يجب تحديدها بواسطة شركة التوزيع قبل الطرح.



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1. SCOPE

This specification defines the minimum technical requirements for design, engineering manufacturing, testing, inspection and performance for medium voltage 12/24 kV, 50 HZ current-limiting fuses (high rupture capacity fuses) Designed for both indoor and outdoor uses intended to be used for protection of distribution transformers and shunt power capacitors from thermal and dynamic effects of short circuits and over currents. One of the major advantages of fuses over circuit breaker protection is their ability to limit the rms and peak values of the prospective short circuit current immediately downstream at the point of installation. Each fuse should be delivered to withstand effect of humidity and occasional condensation

2. Applicable codes and standards:

The fuses should be complying with the latest edition of IEC publications unless otherwise specified in this specification. The equipment/material covered in this specification shall comply with the latest edition/amendment of the following codes and standards. Where any provision of this specification differs from those of the standards listed below, the provisions of this specification shall apply.

Table 1: list of applicable standards

Standards	Title
IEC 60282-1	High-voltage fuses - Part 1: Current-limiting fuses
IEC 62271-105	High-voltage switchgear and control gear – Part 105: Alternating current switch-fuse combinations for rated voltages above 1 kV up to and including 52 KV
IEC/TR 62655	Tutorial and application guide for high-voltage fuses
IEC 60549	High-voltage fuses for the external protection of shunt power capacitors
DIN43625	High-voltage fuses , rated voltages 3.6 to 36kV; fuse-links

3. Environmental conditions :

3.1. Climatic conditions :

- Maximum ambient air temperature in shade 45 °C (50 °C as option)
- Minimum ambient air temperature - 5°C
- Maximum wind velocity at small height 35 m / sec



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- Minimum altitude should not exceed 1000 m
- relative humidity of ambient air of 95%
- Occasional sand storms, salty air, rapid Temperature, Changes and few rains.

4. Design Criteria and Features Required

- 1- The fuse element should be made of high-grade silver – plated copper not less than 4 micron and wound around ceramic cylinder which are embedded in quartz for arc extinction (based on Ampere rating) and a striker pin mechanism.
- 2- In case of any fault, the striker pin comes out with pre-defined force. (This striker pin furthermore can be used to trip equipment it has to protect).
- 3- All fuse links should be equipped with thermal protection, integrated with striker pin mechanism, which complies with requirements specified in IEC 60282-1. The thermal protection is tripping device which is integrated with the striker of HV fuses. It is activated when the allowable temperature is exceeded, and releases the striker which can directly trip the switch-disconnector to open the circuit and avoids further temperature increases. The high temperatures inside the switchgear interior may be caused by external conditions or by a high current passing through the fuse link
- 4- The direction of striker pin operation should be clearly and indelibly marked on the Fuse-Link
- 5- The conductor bands of the fuse element are made of copper coated silver in a spiral, embedded in quartz sand filling and totally sealed within high thermally resistive porcelain housing.
- 6- Fuse selection depends on the maximum load current, type of load, prospective fault current, system voltage and ambient temperature of the installation.
- 7- In a 3-phase installation, it is assumed that all three fuses are subjected to the same rate of degradation.
- 8- The fuse element should be able to carry the rated current continuously at 40°C ambient temperature without exceeding the limit of temperature rise according to IEC60282-1, table 6.
- 9- The fuse-link and the fuse-base should be able to carry their rated current continuously without exceeding the limits of temperature rise given in Table 6, clause4.7 in IEC60282-1and without deterioration.
- 10- The fuse-link contact and fuse-carrier contact should be designed to carry a current up to 400 A.



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- 11- The fuse-link contact, fuse – carrier contact should be made of high-grade silver – plated copper not less than 4 micron.
- 12- The fuse – link body should be made of porcelain.
- 13- The fuse link should be form the fast type (The fuse link should be back-up class).
- 14- Time - current characteristics should be in compliance with IEC 60282-1 standards and should be offered with tender.
- 15- The values of cut – off current corresponding to each value of prospective breaking current should not exceed the values corresponding to the cut – off characteristics given by the manufacturer.
- 16- The manufacturer should indicate the upper limit of the cut-off current corresponding to each value of prospective breaking current up to the rated maximum breaking current of the fuse under specified conditions determined as part of the breaking type tests specified in clause 6.6 in IEC60282-1.
- 17- The fuse manufacturer should define the maximum body temperature that may be reached with any current above the minimum melting current, and the corresponding current value.
- 18- The manufacture should indicate the maximum value of the switching voltage according to IEC 60282-1
- 19- A powder-filled fuse-link should not emit flame or powder, although a minor emission of flame from a striker or indicating device is permissible, providing this should not cause breakdown or significant electrical leakage to earth.
- 20- On each fuse link the badge ofEDC should be clearly and indelibly marked.
- 21- The fuse links should have the dimensions as per attached specification.
- 22- After the fuse being operated, the components of the fuse apart from these intended to be replaced after each operation should be in the original state. It should be possible to remove the fuse-link in one piece after operation.
- 23- After operation, the fuse should be capable of withstanding the power frequency recovery voltage across its terminals.
- 24- Operation should not generate switching voltages higher than the values specified in Table7-caluse 4.9 according to IEC60282-1

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- 25- Standard values of rated TRV are specified in tables 9-IEC60282-1. These values apply to the rated maximum breaking current of a fuse.
- 26- The high rupture capacity fuses should be interrupts the short - circuit current long before its maximum value is attained. This cut - off effect greatly reduce both thermal and magnetic stresses on the equipment protected by the HRC fuse.
- 27- The fuses should be employ fabricated element made of high-grade silver – plated copper not less than 4 micron which are specially designed to give a low temperature rise when carrying their full rated current.

5. Technical Specifications For 11 KV & 22 KV Fuses :

The fuse should have the following technical specification:

Table (1)

Item No.	Specification item		
1	Nominal operating voltage KV	11	22
2	Maximum operating voltage KV	12	24
3	Rated current of fuse element at 40 °C (Ir) A	6.3,10,16,20,25,40,50,63,80, 100,125,160,200A	
4	Minimum melting current of fuse element A	About 3 Ir	About 3 Ir
5	The fuse – link contacts and fuse – carrier contacts should be capable to carry current of A	400	400
6	One – second thermal withstand current for fuse-link contacts and fuse-carrier contacts KA	25	20
7	Dynamic withstand current of fuse-link contacts and fuse-carrier contacts. KA	62.5	40
8	Rated breaking current at nominal operating voltage KA	40	40
9	One minute power frequency withstand voltage(dry and wet) - To earth and between poles - Across isolating distance of fuse base(KV)	28 32	50 60
10	Lightning impulse withstand voltage - To earth and between poles - Across isolating distance of fuse base(KV).rms	75 85	125 145
11	Leakage path of fuse base insulator should be as follows: - for indoor - for outdoor	2 cm/kv 4 cm/kv	

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6. Rated current of the fuse base

- The current assigned to a fuse-base that a fuse-base should carry continuously without exceeding specified temperature rises, when equipped with a fuse-link of the same current rating designed to be used in the particular fuse-base connected to the circuit with certain specified conductor sizes and lengths, at an ambient air temperature of not more than 45°C (50 °C as option).
- The rated current of the fuse-base should be selected from the following values: 200 A(outdoor); 400 A(indoor)...according to IEC60282-1-clause 4.4.

7. Quality and materials :

All materials should be of the best quality and of the suitable Class to meet the working and climatic condition taking into consideration IP 65 & IP 41 at least for outdoor and indoor installation respectively.

8. MARKING

Each fuse or fuse-link specified herewith should be marked of the following minimum information according to clause 5.2 –IEC60282-1:

- a- Manufacture Name or Trademark.
- b- Customer name.
- c- Voltage Rating.
- d- Rated current (I_n) A.
- e- Type of Fuse or Fuse – Link.
- f- Rated Maximum Breaking Current (I_1) A.
- g- Rated Minimum Breaking current (I_3) A.
- h- Manufacturer's type designation.
- i- Year of Manufacture.
- j- Class (back-up, general-purpose, full-range).
- k- Type of striker (heavy), if any.
- l- Location of the striker mechanism (if applicable).
- m- Minimum melting current of fuse element A.



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- n- Maximum application temperature (for fuse-links designed for use in surrounding temperatures above 45 °C(50°C as option) tested in accordance with Annex E in IEC60282-1)
- o- Country of manufacture.

9. Inspection and testing :

- Electricity Distribution Company (...EDC) reserves its right to carry out inspection during fabrication stages and witnessed the testing of fuses
- The methods of carrying out the tests, selection of samples used for tests and rejection of fuses should be accordance with all relevant applicable IEC standards.
- All test should be carried out by the manufacturer and on his expense under control and supervision of (...EDC) inspection engineer (s)
- Three copies for both type and special test certificates should be delivered to (...EDC)
- Attendance of (...EDC) representative (s) during fabrication stages and testing should not relieve the manufacture of his responsibility for furnishing the fuses in accordance with the requirements of this specification and should not give him the right to invalidate any claim that (...EDC) may make because of defective or unsatisfactory material or faulty workmanship
- Type tests certificates should be offered with tender.
- Tests should be made in accordance with IEC 60282-1 and its amendments and may be witnessed by (...EDC) representative.

9.1. TYPE TESTS :

- Conditions for making the tests according to IEC60282-1 Clause (6)
- The type tests to be conducted upon completion of a design or following a change that affects the performance are the following:
 - a- Temperature -rise tests and power-dissipation measurement
 - b- Breaking tests.
 - c- Test for time - current characteristics.
 - d- Test of striker.
 - e- Dielectric tests (fuse-base only).

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9.2. SPECIAL TESTS

- The special test samples should be randomly selected by...EDC. from each offered lot of the same type and ratings The tests should be done at on international well known and approved testing laboratory or manufacturing laboratory as should be mutually agreed upon
- The special tests should thermal shock tests for outdoor fuses.
- Special tests are made to check whether a type or particular design of fuse corresponds to the characteristics specified and behaves satisfactorily under special specified conditions. They are made on samples to check the specified characteristics of all fuses of the same type.
- These tests should be repeated only if the construction is changed in a way which might modify its behavior.

List of special tests

- The following tests are to be made after agreement between manufacturer and user for certain types of fuses or for special applications:
 - Thermal shock tests (for fuses intended to be used outdoors)
 - Power dissipation tests (for fuses not intended for use in enclosures)
 - Waterproof test (ingress of moisture) for fuses intended to be used outdoors
 - Pre-arcing temperature rise and arcing duration withstand tests for back-up fuses for use in switch-fuse combination.

10. Catalogues , Drawings , and samples :

- The tendered should submit with his offer a complete copy of outline drawings, with all dimensions and a complete catalogue containing full technical data and time - current characteristics curves and cut – off characteristics curves of the fuses. He should also submit quality control procedure and origin certificate and test certificates.
- The tenderer should submit with his offer (one) fuse – link for each rating free of charge for purpose of analysis, in case of the offer which should be not accompanied with the above mentioned catalogues and drawings should not be taken into Consideration.

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11. Mechanical characteristics of strikers:

Strikers may be classified according to the amount of energy they are able to deliver to a mechanical switching device or a signaling device between two points A and B (see Figure 1) of their travel and by a minimum withstand force.

The withstand force is the characteristic which prevents the return of the striker, after operation, to less than the minimum actual travel OB when a static external force applied. The mechanical characteristics of the striker are given in

Table (2)

Type	Energy	Mechanical characteristics					
		Value of		Actual travel		Minimum withstand force	Maximum duration of travel (see note)
		Free travel (OA)	Further travel during which energy must be delivered (AB)	Min (OB)	Max (OC)		
	J	mm	mm	mm	mm	N	ms
Medium	1± 0.5	4	16	20	40	20	100
Heavy	2±1	4	6	10	16	40	100

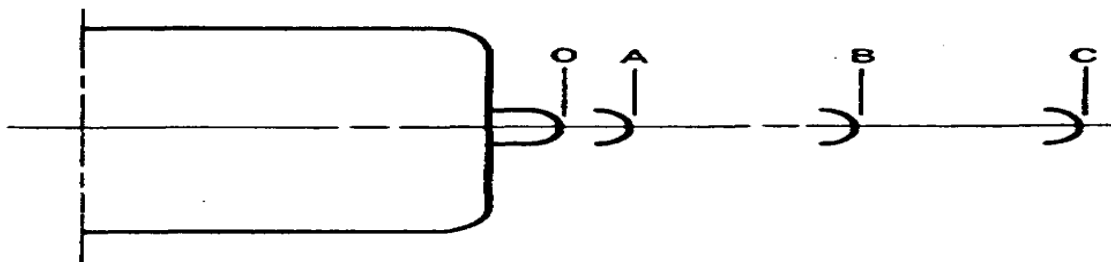


Figure (1) various stages of the striker travel

NOTE: – Duration of travel is defined for actual fuse – links as the time when travel OB is reached. For dummy fuses it is the time from application of voltage until travel OB is reached.

Key

OA Free travel - No energy output specified

AB Further travel during which energy must be delivered

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OB Minimum actual travel

OC Maximum actual travel

CB Maximum permitted return travel under withstand force (when applicable)

NOTE: Duration of travel is defined for actual fuse-links as the time from commencement of arcing to the time when travel OB is reached. For dummy fuses, it is the time from application of voltage until travel OB is reached.

12. Dimensions

12.1. (HIGH RUPTURE FUSE) (H.R.C) dimension FUSES 12 KV

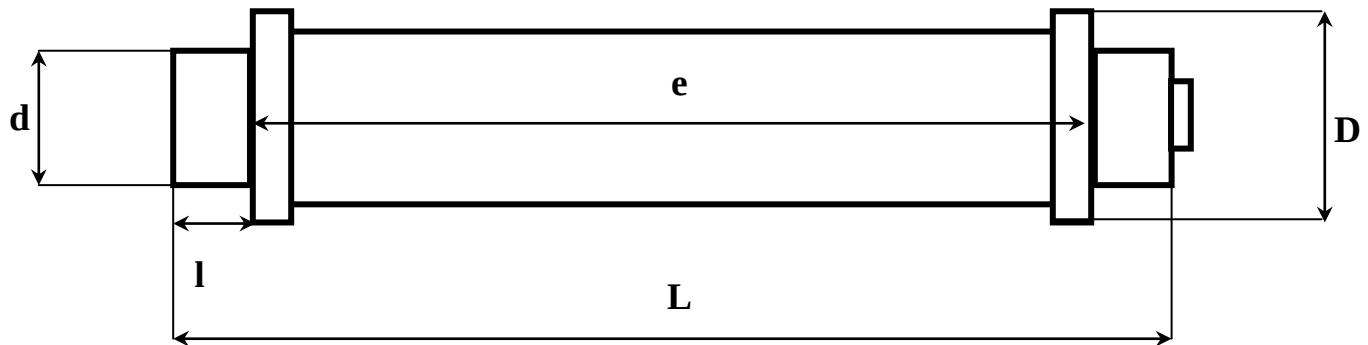


FIG (2)

FUSE LINK RATING	Dimension in mm				
	e	L	l	D	d
12 KV	≈ 292	≈ 360	≈ 35	- 52:62 mm FOR FUSE RATED ≤ 40A - 76:87 mm FOR FUSE RATED ≥ 50A	≈ 45

TABLE (3)

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12.2. (High Rupture FUSE) (H.R.C) dimension Fuses 24 KV:

FUSE LINK RATING	Dimension in mm				
	e	L	l	D	d
24 KV	≈ 442	≈ 508	≈ 35	- 52:62 mm FOR FUSE RATED ≤ 40A - 76:87 mm FOR FUSE RATED ≥ 50A	≈ 45

TABLE (4)

13. Suitable fuses for transformer:

VOLTAGE	The transformer rating KVA	25	50	63	100	160	200	300	500	630	800	1000	1500
12 KV	Nominal current for (H.R.C) fuses	6.3	6.3	10	16	20	25	40	50	63	100	100	160
24KV		6.3	6.3	6.3	10	16	16	25	40	40	40	63	80

TABLE (5)

14. Guarantee:

■ يتم تقديم خطاب اعتماد من الشركة القابضة لكهرباء مصر (ساري)

- The supplier should guarantee the high rupture Fuse against all defects arising out of faulty designed or workmanship or of defective materials for a period of 18 months from putting in service, or 24 months after delivery, whichever expires earlier.
- The good quality and manufacturing of the materials should be guaranteed.
- Any parts which on account of poor quality of the materials, manufacturing defects or poor assembly, should be replaced or repaired in the shortest time possible and free of charge.



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15. Guarantee Tables :

- The tenderer should fill the attached guarantee tables.
- Any tender not accompanied with it clear and complete guarantee tables should be rejected.
- Please, provide explanations for deviations if any.

Guarantee Table For 11&22 KV FUSES

SCHEDULE OF PARTICULARS

EDC Inquiry No: _____ Item No: _____

No.	DESCRIPTION	EDC Specified Values		Vendor Proposed Values
Material, Design and Construction Requirements				
A	Particulars			
1	Type	----		
2	Manufacture’s name	----		
3	Reference manufactured standard	IEC60282-1		
B	Materials			
1	Material of fuse –link contacts	High grade silver – plated copper not less than 4 micron.		
2	Material of fuse –carrier contacts			
3	Material of fuse –element			
C	Dimensions of fuse - link (mm)			
1	Length (mm)	As attachment		
2	Diameter (mm)	As attachment		
3	Type of striker / indicator	Medium at least		
4	Weight (KG)	-----		
5	Insulating body	porcelain		
D	Performance			
1	System Operating Voltage (KV)	11	22	



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2	Maximum Design Voltage (KV)	12	24	
3	Rated Breaking Current at Rated Voltage (KA)	40	40	
4	Melting Time (Second)	According (I-t)curve		
5	Minimum melting current of fuse – element (A)	About 3 Ir		
6	Rated current of fuse element at 40 °C (A)	As required		
7	Current of fuse – link contact (A)	400		
8	Current of fuse – carrier contact (A)	400		
9	One – second of thermal withstand current for (a) fuse-link contacts (KA) (b) fuse-carrier contacts (KA)	25	20	
10	Dynamic withstand current of (a) Fuse-link contacts (KA) (b) Fuse-carrier contacts. (KA)	62.5	40	
11	Transient recovery voltage (KV)	tables 9-IEC60282-1		
12	Maximum temperature rise (°C)	IEC60282-1		
13	Rated power Losses , (Watts)	IEC60282-1		
14	Maximum Duration of Strikers Travel (sec)	IEC60282-1		
15	Maximum Arcing withstand time	Clause 4.15.3-IEC60282-1		
16	Time –Current characteristics	Required		
17	Cut - off characteristics	Required		
18	I^2T characteristics	Required		
19	Pre –arcing Time –Current characteristics	Required		

I / We guarantee the technical data given above for the offered material/equipment.

Signature

Date